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Robust copy-move detection and localization of digital audio based CFCC feature

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Abstract

Copy-move forgery is a common audio tampering technique in which users copy the contents of one speech and paste them into another region of the same speech signal, thus achieving the effect of tampering with the semantics. To verify the authenticity of the audio, this paper proposes a method to detect and localize audio copy-move forgery based on the cochlear filter of cochlear filter cepstral coefficients (CFCC) feature. The pitch tracking algorithm is used to distinguish the voiced and unvoiced segments in the audio, and then the CFCC coefficients are extracted for each voiced segment. The CFCC feature simulates the entire transmission process of signals in the cochlear basilar membrane using wavelet transformation. Finally, we use Pearson correlation coefficients (PCCs) and dynamic time warping (DTW) in combination to compare the similarity of voiced segments, accurately determining the tampered locations in the audio through threshold judgment. Through extensive experiments on relevant datasets, this algorithm achieves a precision rate of 98.39% and a recall rate of 98.00% in detecting audio without post-processing. Even when detecting audio that has undergone different post-processing, the precision and recall rates remain above 90%. Compared to existing methods, this approach not only achieves precise localization of replicated segments but also demonstrates superior experimental results.

Keywords Digital audio · Cochlear filter cepstral coefficient · Copy-move · Pearson correlation coefficients · Robustness

1 Introduction

The rapid development of digital multimedia technology today has made audio, images, videos, and other digital information play an increasingly important role in our lives. These digital media not only bring great convenience but also greatly enrich our daily lives [1, 2]. However, just like any other thing, digital multimedia has its two sides. With the widespread use of multimedia software, some criminals view it as a tool for committing crimes, maliciously tampering with acquired multimedia information to achieve illegal gains. This malicious tampering behavior poses serious harm to society. Therefore, digital multimedia forensics technology has become particularly important. Digital multimedia

Extended author information available on the last page of the article forensics technology includes audio, image, and video forensics techniques. Among them, audio forensics techniques include active and passive digital audio forensics. Active digital audio forensics refers to embedding digital watermarks in original audio files to ensure the authenticity and integrity of the audio and to prevent tampering. However, in practical applications, there are few cases where watermarks are detected in extracted audio. Therefore, passive forensics techniques become more practically valuable. Passive digital audio forensics involves directly analyzing suspicious audio files to determine whether they have been tampered with. The main types of audio tampering include deletion, splicing, and copy-move [3–5]. One common form of manipulation is copy-move audio segments. Copy-move refers to the act of copying a section of the original audio file and pasting it elsewhere within the same audio to alter its content. This type of manipulation can affect the integrity, reliability, and accuracy of the audio file. The user does not need to have extensive technical expertise, as the relevant editing software can be used to perform copy-move operations on audio files. For example, "I am not a thief, he is a thief." can be manipulated by copy-move to become "I am not a thief, he is not a thief."The audio signal tampered through copy-move is shown in Fig. 1, where the two red segments represent the copied and moved manipulated portions. If the tampered audio file undergoes post-processing operations such as adding background noise to eliminate tampering traces, it will become even more difficult to detect [6]. Nowadays, digital audio evidence has become indispensable evidence in court, but the authenticity of the audio is a prerequisite for the evidence. Therefore, developing detection technology for copy-move audio manipulation has become an urgent problem.

In recent years, many scholars have been highly concerned with speech forgery detection, and they have proposed many methods for detecting audio copy-move forgery. For example, Xiao et al. [7] proposed a copy-move forgery detection method based on audio waveform. The audio is first split into equal-length segments, and then a fast convolution algorithm is used to calculate the similarity between the audio segments, which in turn determines whether there are copy-move tampered segments in the tested audio. However, if there is reverberation that changes the waveform, it can severely affect the experimental results. Imran et al. [8] proposed a copy-move forgery detection algorithm using a 1-D local binary pattern operator,

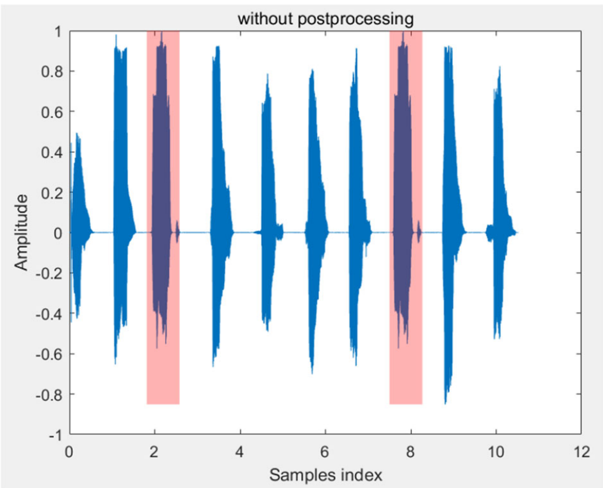

Fig. 1 A forged audio by copy-move forgery

which segments the audio into silent and voiced segments using the voice activity detection method, and then extracts words from the audio, and applies a 1-D local binary pattern operator to obtain the histogram features of these words, then use mean squared error (MSE) to calculate the similarity between any two histograms. The detection accuracy of this method is very dependent on speech activity detection, which can lead to accuracy degradation due to inaccurate boundary point detection. Kucukugurlu et al. [9] improved it by proposing a method based on pitch and 1-D local binary pattern operator. Although this method significantly improves the detection performance of boundary points, its accuracy in detecting post-processed audio is not high. Xie et al. [10] proposed a copy-move forgery detection method based on multi-feature fusion, which uses the C4.5 decision tree to fuse gammatone frequency cepstral coefficient features, Mel frequency cepstral coefficient (MFCC) features, pitch features, and discrete Fourier transform (DFT) coefficient features. Finally, the Pearson correlation coefficient (PCCs) and average difference (AD) are used to determine the detection results. However, the algorithm requires a high time complexity for feature fusion using C4.5 decision trees for the four features. Liu et al. [11] proposed a fast method for detecting copy-move forgery in audio files. Firstly, the audio is divided into syllables then discrete Fourier transform (DFT) is performed and finally the audio segments are sorted. As a result, only the similarity between neighboring audio segments needs to be compared, which reduces the time complexity from $O(n^2)$ to $O(n\log n)$. Li et al. [12] proposed a fast detection algorithm based on pitch features and using a different comparison algorithm, but the algorithm will be more difficult to select due to the threshold value in the similarity comparison, which finally leads to a high false positive rate. Wang et al. [13] proposed a copy-move forgery detection algorithm based on discrete cosine transform (DCT) and singular value decomposition (SVD). This method has significantly improved the detection time, but it is prone to causing false positives. Mannepalli et al. [14] proposed a copy-move forgery detection based on MFCC and zero crossing rate. However, the robustness of the method is low for the post-processed audio. Based on the reference [14], Akdeniz et al. [15] proposed a copy-move detection algorithm based on Mel frequency and Delta-Mel frequency features. However, the detection accuracy of this method will decrease under low signal-to-noise ratio conditions. Yan et al. [16] proposed a copy-move forgery detection method based on pitch similarity. However, this method can achieve better results only in detecting audio without post-processing, but the detection accuracy is low for audio after post-processing. Based on the algorithms in [16], the same authors improved it by proposing a forgery detection method based on pitch and formant [17]. However, this method has a low detection accuracy for some post-processing operations that can change the pitch and formant of the audio signal. Ustubioglu et al. [18] proposed a copy-move forgery detection algorithm based on improved cosine transform, utilizing the euclidean distance (ED) to determine the similarity of each audio segment. However, when using the euclidean distance for similarity comparison, the algorithm exhibits a higher false positive rate, and it lacks robustness when detecting post-processed audio. In recent years, algorithms for detecting copy-move audio forgery based on graph matching have become increasingly popular [19–22]. Ustubioglu et al. [22] proposed a copy-move forgery detection algorithm based on keypoint matching in the Mel spectrogram. Initially, this algorithm extracts SIFT keypoints from the RGB channels of the Mel spectrogram and then performs keypoint matching and localization. However, when detecting audio with low signal-to-noise ratio, the algorithm may encounter keypoint displacement, leading to inaccurate localization. Therefore, it can be observed that the current methods for detecting copy-move forgery based on homologous audio have two issues: 1) the forgery location cannot be accurately determined, and 2) the detection accuracy is low for copy-move forgery audio signal with post-processed.

In order to address the aforementioned issues, we propose a copy-move forgery detection method based on cochlear filter cepstral coefficient (CFCC) features. This method utilizes CFCC feature extracted using a human auditory model, which exhibit high stability even for post-processed audio. Firstly, the voiced segments are extracted from the audio using a pitch tracking algorithm. Then, CFCC coefficients are computed for each speech segment. Finally, we utilize Pearson correlation coefficients and dynamic time warping (DTW) algorithm to jointly compute the similarity between each audio segment. When the Pearson correlation coefficient exceeds the set threshold and the DTW value is below the designated threshold, the audio segment is identified as having undergone copy-move forgery and the tampered portions can be accurately located. Extensive experiments demonstrate the strong robustness of this method. The contribution made by the algorithm proposed in this paper to audio copy-move forgery detection can be summarized as follows:

1. Voice activity detection (VAD) is the process of extracting speech segments from audio files. The algorithm proposed in this paper utilizes a pitch tracking algorithm to achieve this goal. YAAPT is a pitch tracking algorithm known for its high accuracy and robustness. Accurately extracting speech segments is crucial for the precise localization of the algorithm.
2. Cochlear filter cepstral coefficients (CFCC) is a feature parameter extracted based on auditory perception to simulate the auditory process of the human ear. We use the wavelet basis function to model the signal propagation process on the cochlear basilar membrane, and use a nonlinear power function to compress the signal, and finally use the discrete cosine transform for decorrelation. The final CFCC features obtained have high robustness.
3. We use Pearson correlation coefficient (PCCs)and dynamic time warping (DTW) algorithm to jointly determine the similarity between speech features. Compared to a single algorithm, this decision-making approach significantly reduces the false detection rate, thereby improving the detection accuracy of the algorithm.

The second part of the article provides a detailed description of the implementation process of the copy-move forgery detection algorithm, including the extraction process of audio segments and CFCC feature extraction, as well as similarity comparison. The third part elaborates on the experimental results, including the database used, threshold selection, and experimental analysis. The fourth part concludes the entire article.

2 Proposed method

In this section, we provide a detailed description of the copy-move forgery detection process, which consists of three main parts: voice activity detection, feature extraction, and similarity comparison. The flow chart of the overall design of the algorithm is shown in Fig. 2.

2.1 Yet another algorithm of pitch tracking (YAAPT)

Voice activity detection is an important task in speech processing and speech recognition for distinguishing voiced segments from unvoiced segments in speech signals [23]. In this thesis, a voice activity detection method based on the pitch tracking algorithm is adopted [24]. The algorithm is mainly combined with the method of estimating the fundamental frequency in the time-frequency domain. The method achieves voice activity detection through four steps:

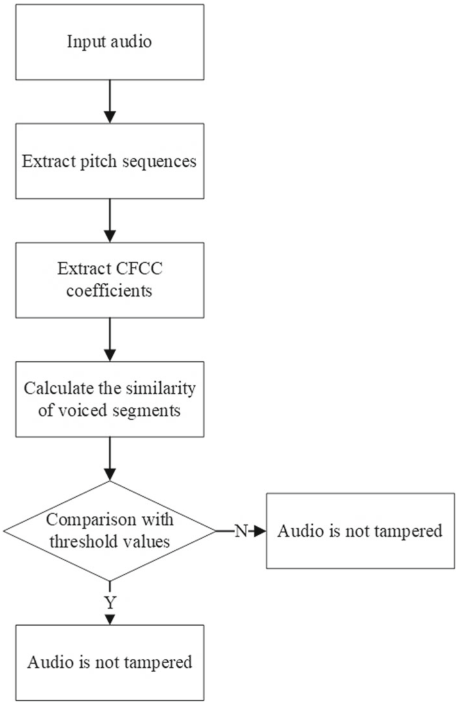

Fig. 2 The copy-move forgery detection process

pre-processing of speech signals, spectral pitch tracking, pitch candidate estimation, and final pitch determination.

- Step 1: Preprocessing
  In the pitch tracking algorithm YAAPT, nonlinear preprocessing is used to create multiple types of signals by performing absolute and squared values on the speech signal. After nonlinear preprocessing can be used in subsequent steps to improve the performance of pitch estimation, for example for partial recovery of missing fundamental frequency information, thus improving the accuracy and robustness of pitch tracking. The suspicious speech signal x(n) is transformed into s(n) through a square nonlinear processing, as shown in (1).

$$s(n) = x(n)^2 \tag{1}$$

- Step 2: Spectral pitch tracking
  SHC is a technique used to estimate approximate pitch trajectories. The formula calculates the correlation between the local harmonic structure of the speech signal in the time domain and the harmonic structure in the frequency domain [25]. SHC is calculated by (2).

$$SHC(i, f) = \sum_{f'=-W/2}^{W/2} \prod_{r=1}^{N+1} S\left(i, rf + f'\right) \tag{2}$$

  Assuming the preprocessed signal is z(n), it is framed and windowed, for each frame is noted as G_i, i=1, 2, ... , N. After performing Fourier transform, we obtain the amplitude spectrum S(i, f). S(i, f) denotes the amplitude spectrum of frame i at frequency f. W and N denote the window length and harmonic number, respectively.

NLFER(t) represents the normalized low-frequency energy ratio of the speech signal. By comparing the ratio of low-frequency energy to full-frequency energy, a larger ratio indicates an audible frame, and vice versa, a silent frame. NLFER is calculated by (3).

$$NLFER(i) = \frac{\sum_{f=2\times F_{0\_\min}}^{F_{0\_\max}} S(i, f)}{\frac{1}{T}\sum_{i=1}^{T}\sum_{f=2\times F_{0\_\min}}^{F_{0\_\max}} S(i, f)} \tag{3}$$

where T denotes the number of frames, $F_{0\_\min}$ denotes the minimum value of spectral energy, and $F_{0\_\max}$ denotes the upper limit of spectral energy.

- Step 3: Temporal pitch tracking based on NCCF
  The candidate pitch sequence is obtained by calculating the NCCF of the original signal and the nonlinearly processed signal. NCCF is calculated by (4).

$$NCCF(k) = \frac{1}{\sqrt{e_0 e_k}} \sum_{n=0}^{L-K\max} s(n)s(n+k) \tag{4}$$

$$e_k = \sum_{n=k}^{n=k+L-K_{\max}} s^2(n) \tag{5}$$

where e_k denotes the energy of the speech signal s(n) in the range from n=k to n=k+L-$K_{\max}$.

$$e_0 = \sum_{n=0}^{n=L-K_{\max}} s^2(n) \tag{6}$$

where e_0 denotes the energy of the speech signal s(n) in the range from n=0 to n=L-$K_{\max}$.

$$K_{\min} \leq k \leq K_{\max} \tag{7}$$

where s(n) denotes the time domain signal of a frame and L denotes the frame length. $K_{\min}$ and $K_{\max}$ are lagged values.

- Step 4: Dynamic Programming
  Through the previous three steps, the pitch candidates are obtained, and we need to process them further to get the final pitch, using the method of dynamic programming. Dynamic programming is an optimization algorithm that finds the optimal sequence among all possible pitch sequences. Finally, iterate through each element of the pitch array and if the value is not 0, it is marked as a voiced segment, otherwise, it is a voiceless segment. The pitch tracking algorithm segments the original audio signal x(n) into k voiced segments $\{y_1, y_2, ..., y_k\}$.

2.2 Cochlear filter cepstral coefficients (CFCC)

Cochlear filter cepstral coefficients (CFCC) is a feature parameter first proposed by Li Q from Bell Labs in 2011 and applied to speaker recognition [26–28]. This feature is extracted based on auditory perception and simulates the auditory process of the human ear. Compared to the more commonly used Mel frequency cepstral coefficients (MFCC) and gammatone frequency cepstral coefficients (GFCC) based on Mel and gammatone filters, respectively, CFCC features have higher recognition accuracy, especially in noisy environments, and have higher robustness [29]. The CFCC features are extracted based on the auditory transform, which then uses the cochlear filter function as a wavelet basis function, and the wavelet transform is used to simulate the entire transmission process of the signal in the cochlear basilar membrane. The feature extraction process of CFCC is shown in Fig. 3.

- Step 1: Pre-emphasis
  Pre-emphasis is a common preprocessing step in speech processing, which usually involves using a high-pass filter to obtain the high-frequency components of a speech signal. This can help to extract more information from the speech signal, better distinguish high-frequency peaks, and improve the intelligibility and reliability of the speech signal. Pre-emphasis enhances the high-frequency components of the audio signal, not only making the energy distribution more even in the high-frequency range but also increasing the signal-to-noise ratio, thereby making it easier to distinguish between the audio signal and the noise signal. The pre-emphasis transfer function is given by (8).

$$H(z) = 1 - \mu z^{-1} \tag{8}$$

  where μ is a constant, $0.9<\mu<1$. The voiced segment of the speech signal extracted using the pitch tracking algorithm is represented as y(t), and after pre-emphasis, it becomes f(t).
- Step 2: Auditory Transform
  Auditory transform is a signal processing method that transforms sound signals into signals that resemble those heard by human ears. It can make the frequency spectrum of speech signals more consistent with the perceptual characteristics of human hearing while suppressing noise and non-speech components. Auditory transform uses the cochlear filter bank as the wavelet basis function and then filters the signal using wavelet transform. It is a new method for processing speech signals. The original speech signal is f(t), and the output result is T(a,b) after auditory transformation. The T(a,b) is defined as follows:

$$T(a, b) = \int_{-\infty}^{+\infty} f(t)\varphi_{a,b}(t)\mathrm{d}t \tag{9}$$

  in which the $\varphi_{a,b}$ is defined as follows:

$$\varphi_{a,b} = \frac{1}{\sqrt{a}}\frac{(t-b)^{\alpha}}{a}\exp\left[-2\pi f_{\mathrm{L}}\beta\left(\frac{t-b}{a}\right)\right]\cdot\cos\left[2\pi f_{\mathrm{L}}\left(\frac{t-b}{a}+\theta\right)\right]u(t-b) \tag{10}$$

$$a = \frac{f_L}{f_c} \tag{11}$$

  where α and β determine the time-frequency shape and width of $\varphi_{a,b}(t)$. In this experiment, α is set to 3, and β is set to 0.2 because this combination yields the best noise reduction effect. θ is the initial phase, u(x) is the unit step function, f_L is the lowest center frequency of the cochlear filter bank, and f_c is the current center frequency of the cochlear filter. The cochlear filter was analyzed using the Bark scale for the center frequency distribution, and we used the 16-channel Bark scale with center frequency f_c = [250 350 450 570 700 840 1000 1170 1370 1600 1850 2150 2500 2900 3400 4000], a is the scaling factor, and b is the displacement factor and is set to 0.
- Step 3: Hair Cell Window
  The cochlear window is a window function similar to the hair cells in the human ear, which can be used to further enhance the feature discrimination of speech signals in the time domain. We can simulate the action of the hair cells through (12).

$$h(a, b) = [T(a, b)]^2 \tag{12}$$

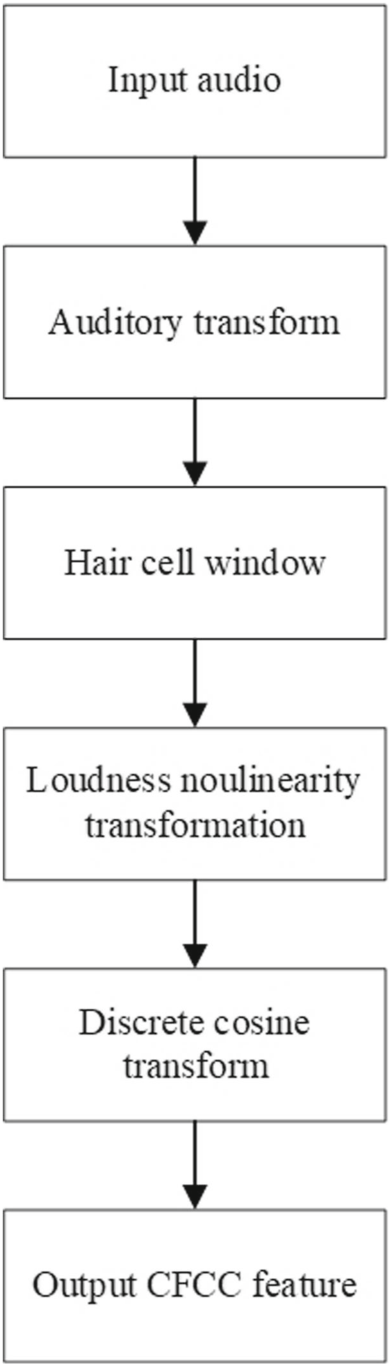

Fig. 3 The process of CFCC feature extracted

where T(a,b) the output of the cochlear filter.

The energy information of the current signal h(a,b) can be obtained through (12), and then after the window function transformation can be obtained for each hair cell output. The size of the cochlear window can be adjusted according to specific needs. Generally speaking, the smaller the window size, the finer the feature information obtained, but it may also lead to a more complex analysis process and higher computational complexity. In experiments, a variable window length is used for smoothing processing because the response time of the auditory nerve to sound decreases with increasing frequency. The specific calculation of the hair cell window function is shown in (13).

$$S(i,j)=\frac{1}{d}\sum_{b=l}^{l+d-1}h(i,b),l=1,L,2L,\cdots; \tag{13}$$

$$d_i=\begin{cases}3.5\tau_i & 3.5\tau_i<20\text{ ms}\\ 20\text{ ms} & 3.5\tau_i\geq 20\text{ ms}\end{cases} \tag{14}$$

$$\tau_i=1/f_c \tag{15}$$

where d_i denotes the variable window length, τ_i denotes the center frequency period of the i-th filter, and L denotes the frame shift.

- Step 4: Loudness Noulinearity Transformation (LNT)
  After the output through the hair cell window, a cubic root transformation function is used for nonlinear loudness transformation.

$$y(i,j)=[S(i,j)]^{\frac{1}{3}} \tag{16}$$

- Step 5: Discrete Cosine Transform (DCT)

$$CFCC(i,n)=\sqrt{\frac{2}{M}}\sum_{m=1}^{M-1}y(i,m)\cos(\frac{\pi n(m-\frac{1}{2})}{M} \tag{17}$$

where i represents the number of frames and n represents the number of filters.

2.3 Pearson correlation coefficient (PCCs)

The Pearson correlation coefficient is a statistical measure used to quantify the linear correlation between two variables, usually denoted as “r”. The Pearson correlation coefficient range [-1,1], where -1 represents a perfect negative correlation, 1 represents a perfect positive correlation, and 0 represents no linear correlation. Assuming 2 vectors x_i and y_i, the Pearson correlation coefficients of x_i and y_i are calculated by (18).

$$r=\frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}} \tag{18}$$

2.4 Dynamic time warping (DTW)

The dynamic time warping (DTW) algorithm is an effective method for aligning time series, specifically addressing issues such as varying sequence lengths and time axis misalignments. DTW accurately measures the similarity between sequences, providing a reliable means of similarity measurement. Therefore, we utilize the DTW algorithm to evaluate the similarity between different audio segments. The DTW computation proceeds as follows: assuming the two feature matrices are denoted as X and Y. The length of feature matrix X is m, and the length of feature matrix Y is n.

$$X = \begin{bmatrix} x_{1,1} & x_{1,1} & \cdots & x_{1,m} \\ x_{2,1} & x_{2,1} & \cdots & x_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{k,1} & x_{k,1} & \cdots & x_{k,m} \end{bmatrix} \tag{19}$$

$$Y = \begin{bmatrix} y_{1,1} & y_{1,1} & \cdots & y_{1,n} \\ y_{2,1} & y_{2,1} & \cdots & y_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{k,1} & y_{k,1} & \cdots & y_{k,n} \end{bmatrix} \tag{20}$$

$d_{mn}(X, Y)$ denotes the distance between the m-th sample in the X-feature matrix and the n-th sample in the Y-feature matrix, and $d_{mn}(X, Y)$ is calculated as shown in (21).

$$dmn(X, Y) = \sqrt{\sum_{k=1}^{K} \left(x_{k,m} - y_{k,n}\right) \times \left(x_{k,m} - y_{k,n}\right)} \tag{21}$$

$$D = \begin{bmatrix} d_{1,N} & d_{2,N} & \cdots & d_{M,N} \\ \vdots & \vdots & \ddots & \vdots \\ d_{1,2} & d_{2,2} & \cdots & d_{M,2} \\ d_{1,1} & d_{2,1} & \cdots & d_{M,1} \end{bmatrix} \tag{22}$$

We will find a single shortest path in the matrix D starting at $d_{11}(X, Y)$ and ending at $d_{mn}(X, Y)$, and compare the similarity between the audio clips by comparing the DTW distances between the feature matrices, and the path length d is shown in (23).

$$d = \sum_{m \in ix, n \in iy} d_{mn}(X, Y) \tag{23}$$

3 Experimental results

In this section, we first introduce the audio database used in this experiment in Section 3.1. Subsequently, we provide a detailed description of the threshold selection process in Section 3.2 and present a specific detection example in Section 3.3. Following that, in Section 3.4, we introduce the performance metrics employed. Finally, in Section 3.5, we showcase the experimental results, and in Section 3.6, we compare the proposed algorithm with other copy-move forgery detection algorithms.

3.1 Copy-move forged dataset

The copy-move forgery database used in this experiment was generated using the TIMIT database [18]. It consists of 250 audio files, with an audio sampling rate of 8kHz, and each file has a duration of approximately 4 seconds. The audio format is WAV. For each audio file, we randomly select 1 voiced segments and paste them into different positions within the

Table 1 Statistical result of PCCs

PCCs	Over 0.98	Over 0.96	Over 0.90	Over 0.85
Duplicated	88.40%	92.80%	99.20%	99.60%
Normal	0%	2.00%	14.00%	19.2%

same audio. The tampered sound segments are approximately 0.2 to 0.6 seconds in duration. In the end, the database generated based on vad contains 148 forged audio samples, while the one based on pitch contains 368 forged audio samples. We selected 125 samples from each of these two databases to create a mixed database. To simulate realistic copy-move forgery scenarios, we applied some common post-processing operations to the generated forged audio files, including adding noise, median filtering, and MP3 compressing. Gaussian white noise with levels of 30dB, 20dB, and 10dB was added to the attacked audio files to generate a noisy speech database. Additionally, each attacked audio was processed with a median filter. We also applied MP3 compression at 32kbps and 64kbps to the copy-move tampered audio. In total, these 250 copy-move forged audio files were used to generate 1500 post-processed audio files. The configuration environment used in this experiment is as follows: Intel(R) Core(TM) i5-12400F 2.50 GHz, 32 GB DDR4 RAM, Windows 10 operating system, and MATLAB 2022a software as the experimental platform.

3.2 Threshold selection

To enhance the accuracy of the algorithm, appropriate thresholds play a crucial role. In this study, we utilize the joint assessment of PCCs and DTW algorithms to evaluate the similarity between voiced segments, thereby reducing false positive rates. Table 1 presents the statistical results of PCCs, where, with a threshold set at 0.9, copied audio segments account for 99.20%, while normal audio segments account for 14.00%. Table 2 illustrates the statistical results of DTW values, where, with a threshold set at 50, copied audio segments represent 97.60%, and normal audio segments account for 2.4%. Ultimately, we define that when the PCCs value exceeds 0.9 and the DTW value is less than 50, two speech segments are identified as duplicates.

3.3 Detection example

In this section, we will explain how the algorithm detects copy-move forgery through an example. First, we extract voiced segments using the pitch tracking algorithm, as shown in Figs. 5 and 4. The solid red lines indicate the start positions of each voiced segment, and the dashed red lines indicate the end positions of each voiced segment. Then we extract the CFCC feature coefficients for each voiced segment and compare the similarity between them. In the sample audio, we can observe that we have extracted 5 voiced segments and compared their similarity using Pearson correlation coefficient, as shown in Table 3. The Pearson

Table 2 Statistical result of DTW

DTW	0–20	0–40	0–50	0–100
Duplicated	90.10%	93.60%	97.60%	98.80%
Normal	0%	0%	2.4%	14.4%

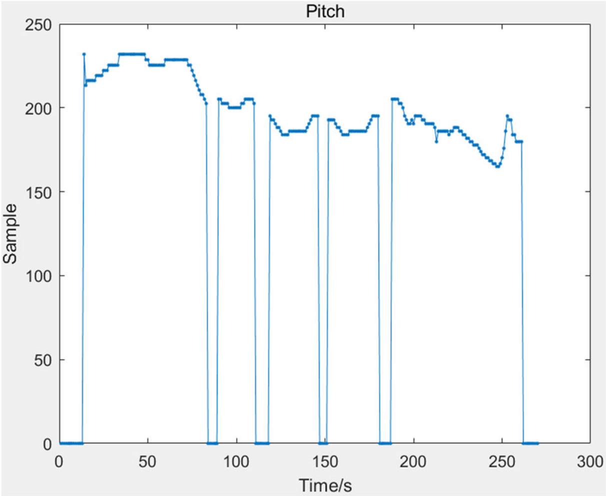

Fig. 4 Pitch sequences of the audio

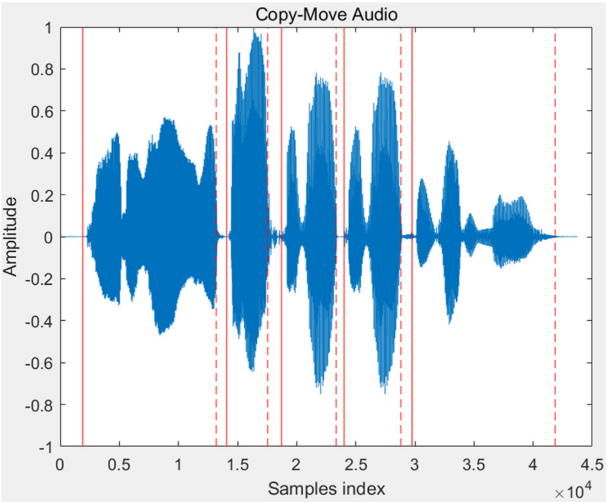

Fig. 5 The copy-move forgery detection

Table 3 PCCs of copy-move audio

Segments	1	2	3	4	5
1	1	0.4096	0.4187	0.4192	0.7521
2	0.4096	1	0.6566	0.6230	0.5148
3	0.4187	0.6566	1	0.9923	0.7422
4	0.4192	0.6230	0.9923	1	0.7402
5	0.7521	0.5148	0.7422	0.7402	1

Table 4 DTW of copy-move audio

Segments	1	2	3	4	5
1	0	178.7155	162.0221	164.1298	223.2051
2	178.7155	0	53.8549	56.9010	223.8906
3	162.0221	53.8549	0	14.9158	138.8489
4	164.1298	56.9010	14.9158	0	141.5771
5	223.2051	223.8906	138.8487	141.5771	0

correlation coefficient between the third and fourth voiced segments is 0.9923, exceeding the set threshold of 0.9. Similarly, we apply the DTW algorithm to calculate the similarity between each voiced segment. The specific results are shown in Table 4. The DTW value between the third and fourth voiced segments is 14.9158, which is below the set threshold of 50. Thus, we determine that there is a copy-move relationship between these two voiced segments, identifying them as tampered audio.

3.4 Performance metrics

In order to verify the robustness of this feature, two performance indicators are used for evaluation, namely: precision, and recall.

Precision is defined by (24):

$$precision = \frac{TP}{TP + FP} \tag{24}$$

Recall is defined by (25):

$$recall = \frac{TP}{TP + FN} \tag{25}$$

where TP represents the number of correctly detected tampered audio samples, FP represents the number of wrongly detected untampered audio samples, and FN represents the number of undetected tampered audio samples.

3.5 Experimental results and analysis

The results of the audio detection are given in Table 5. Without any post-processing, the precision of the audio detection is 98.39% and the recall is 98.00.%. When 30 dB Gaussian

Table 5 Detection result

Method	The method	
Type of attack	Precision	Recall
No attack	**98.39%**	**98.00%**
30dB Gaussian noise	**97.18%**	**96.40%**
20 dB Gaussian noise	**95.93%**	**94.40%**
10 dB Gaussian noise	**93.47%**	**91.60%**
Median filter	**96.76%**	**95.60%**
MP3 (32 kbps)	**95.94%**	**94.80%**
MP3 (64 Kbps)	**97.19%**	**96.80%**

Table 6 Statistical result of precision

Method	Yan [17]	Ustubioglu [18]	Wang [13]	Ustubioglu [22]	The method
Type of attack	Precision	Precision	Precision	Precision	Precision
No attack	93.09%	90.25%	87.50%	91.70%	**98.39%**
30dB Gaussian noise	89.75%	84.23%	84.68%	87.08%	**97.18%**
20dB Gaussian noise	86.72%	81.07%	78.33%	78.33%	**95.93%**
10dB Gaussian noise	82.57%	78.35%	73.18%	74.89%	**93.47%**
Median filter	85.31%	87.65%	79.57%	83.86%	**96.76%**
MP3 (32 kbps)	81.89%	82.38%	75.42%	88.94%	**95.97%**
MP3 (62 kbps)	86.83%	85.66%	77.02%	83.06%	**97.19%**

white noise is added to the audio, the detection precision is 97.18% and the recall rate is 96.40%. When 20 dB Gaussian white noise is added to the audio, the detection precision is 95.93% and the recall rate is 94.40%. When 10 dB Gaussian white noise is added to the audio, the detection precision is 93.47% and the recall rate is 91.60%. When the median filtering attack is applied to the audio, the precisionof detection is 96.76% and the recall is 95.60%. When the audio is compressed in MP3 format and the compression ratio is 32kbps, the precision of detection is 95.94% and the recall is 94.80%. When the audio was compressed in MP3 format and the compression ratio was 64kbps, the detection precision was 97.19% and the recall rate was 96.80%. From the table, it can be seen that after different post-processing attacks on the detected audio, it can still achieve a great detection effect, which proves that the method proposed has a high precision and robustness.

3.6 Comparisons with other methods

To verify the performance of the algorithm, we compared it with several state-of-the-art methods, including those proposed by Yan et al. [17], Ustubioglu et al. [18], Wang et al. [13], and Ustubioglu et al. [22]. Tables 6 and 7 present the comparison results of the proposed algorithm with four other state-of-the-art algorithms in terms of precision and recall. The present algorithm has high precision and recall both under noise addition, recompression and median filtering attacks, especially in the noise case, the CFCC features have high interference resistance. The experimental data show that our proposed audio copy-move forgery detection algorithm based on CFCC features has high precision and robustness.

Table 7 Statistical result of recall

Method	Yan [17]	Ustubioglu [18]	Wang [13]	Ustubioglu [22]	The method
Type of attack	Recall	Recall	Recall	Recall	Recall
No attack	91.60%	85.20%	84.00%	88.40%	**98.00%**
30dB Gaussian noise	87.60%	81.20%	79.60%	83.60%	**96.40%**
20dB Gaussian noise	83.60%	78.80%	75.20%	75.20%	**94.40%**
10dB Gaussian noise	79.60%	72.40%	64.40%	70.40%	**91.60%**
Median filter	83.60%	85.20%	74.80%	78.80%	**95.60%**
MP3 (32 kbps)	79.60%	80.40%	68.40%	84.40%	**94.80%**
MP3 (62 kbps)	84.40%	83.60%	72.40%	81.60%	**96.80%**

4 Conclusion

In this paper, we propose an audio copy-move forgery detection algorithm based on CFCC features. The algorithm utilizes the YAAPT algorithm to extract voiced segments from the audio, and compares the similarity of CFCC features between these segments using Pearson correlation coefficient and dynamic time warping algorithm. Through extensive experimentation on a large dataset of post-processed audio samples and comparison with other state-of-the-art literature, the experimental results demonstrate that this method achieves higher precision and robustness. In future work, we intend to enhance the precision of the algorithm by improving the CFCC features or integrating multiple features to extract more resilient speech features.

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Data Availability Statement The data that support the findings of this study are available from the corresponding author on reasonable request.

Declarations

Conflict of Interest The authors declare that they have no known financial interests or personal relationships that might influence the work reported in this paper.

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